Do Now · retrieval

Lesson 049 · Stack<T>: LIFO and Program State

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

How does List<T>.RemoveAt differ from Remove?

Q2

What is a stack's removal order?

Q3

What is an instance method?

Q4

An empty stack executes push(A), push(B), push(C), x=pop(), push(D), push(x), then pop(). Trace the final stack from bottom to top.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — A stack is last-in, first-out (LIFO). Push places a value on top, Pop removes the top, and Peek reads without removing.

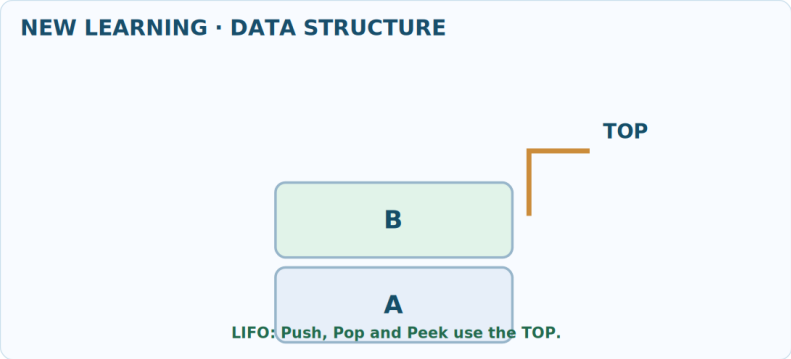
Core idea 2

Explain — A stack can support undo histories and recursive call frames. The physical call stack is a related example but not this Stack<T> collection.

Core idea 3

Apply — Always check Count or otherwise handle an empty-stack exception before Pop/Peek.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

LIFO — Last In First Out.

Push — Add to top of stack.

Pop — Remove and return top.

Peek — View top without removing.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

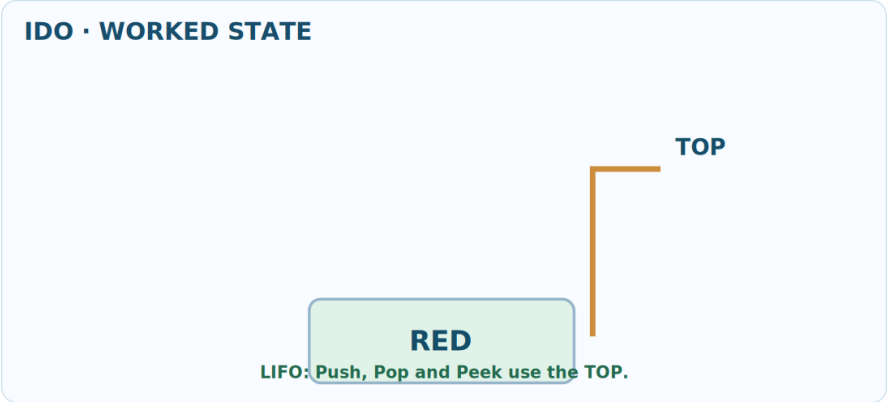
Problem

Trace stack push RED, push BLUE, pop.

Follow the worked method

1. Trace and justify — Popped BLUE; remaining stack top RED. Most recent value leaves first.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Trace and justify
Stack bottom-to-top [2,9], Peek returns 9; popped value was 5.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Push A, B, C; Pop once, then Peek. State both outputs and remaining stack. [6]

Question 2

Write C# to push 3,6,9 and safely pop one value. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why stack suits undo history but queue suits printing jobs; include two operations. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Peek permanently removes the last value from a stack.

Use key vocabulary from memory

1. Explain LIFO in your own words.

2. Explain Push in your own words.

3. Explain Pop in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.